

Project Design Document

Overview

This chatbot web app is built as a browser-based interface with a Node.js backend. It supports local AI execution through Ollama, browser-based saved chat management, and export capabilities.

Architecture

- **Frontend:** `index.html`, `style.css`, `script.js`
 - **Backend:** `server.js` using Express
 - **Storage:** Browser `localStorage` for saved chats
 - **AI Integration:** Ollama local model accessed via `/chat`
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Components

Frontend

- **Chat window:** displays conversation messages
- **Input area:** allows user text entry and submit
- **Saved chats sidebar:** lists saved conversations
- **Buttons:** Send, Save Chat, Rename, Load, Delete, Export
- **Typing indicator:** shows status when AI is processing

Backend

- **Express server:** handles POST `/chat`
 - **Environment config:** `.env` controls `USE_OLLAMA`, `OLLAMA_MODEL`, and `OLLAMA_PORT`
 - **Response processing:** reads streamed Ollama JSON lines and concatenates replies
 - **Error handling:** returns friendly error replies when backend fails
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Data Flow

1. User types a message and sends it.
 2. Frontend posts the message to `/chat`.
 3. Backend forwards the prompt to Ollama and reads streaming results.
 4. The chat response is returned to the client.
 5. The frontend appends the bot message to the chat window.
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Storage Design

- Chats are stored as JSON objects in `localStorage` under `savedChats`.
- Each saved chat includes:
 - `title`
 - `timestamp`

- `messages` array
 - Message entries include:
 - `sender`
 - `text`
 - `time`
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UI Design Principles

- Dark theme for modern readability
- Clear distinction between user and AI messages
- Responsive layout with sidebar and main chat area
- Minimal button labels and consistent spacing
- Accessible message text and simple controls

Technical Requirements

- Run the Node server locally with `node server.js`
 - Ensure Ollama is installed and running on the configured port
 - Use the browser to open `index.html`
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Security and Reliability

- Secrets are kept out of source control via `.env`
- The app degrades gracefully when the AI backend is unavailable
- Local storage persists chat history without server dependencies

Future Enhancements

- Add conversation search within saved chat history
- Support Markdown rendering in chat messages
- Add voice input and text-to-speech
- Add cloud backup for saved chats